

Assignment 3

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Chapter 6

P1

For generator $G = 10011$, the CRC degree is $r = 5 - 1 = 4$. So we divide D with 4 appended zeros: $D = 10101010100000$. After performing modulo-2 division of D by 10011 gives the remainder $R = 0100$, and the remainder length is $|R| = 4$ bits.

P15

a.

No. E first checks the destination IP (i.e., F)'s subnet prefix and realizes that F is in the same LAN. Therefore, E does not send the packet to the default router R1, it sends it directly on the local link.

Source IP : E's IP address. (1)

Destination IP : F's IP address. (2)

Source MAC : E's MAC address. (3)

Destination MAC : F's MAC address. (4)

b.

No. E determines that B is not in Subnet 3, so it will not ARP for B's MAC address. Instead, E must forward the IP datagram to its default gateway, i.e., R1's interface on Subnet 3.

In the Ethernet frame from E to R1:

Source IP : E's IP address. (5)

Destination IP : B's IP address. (6)

Source MAC : E's MAC address. (7)

Destination MAC : The MAC address of R1's interface connecting to Subnet 3. (8)

c.

When S1 receives this ARP Request, it will: (1) learn that A_MAC is reachable via the incoming port (toward Subnet 1) and adds/updates the entry for A in its forwarding table, and (2) forwards the ARP Request out all ports except the incoming port (a broadcast since it is a broadcast frame).

Yes, R1 will receive it. But no, R1 won't forward it to Subnet 3.

B won't send ARP query message asking for A's MAC. Because this ARP Request already includes the sender information and B can use that information directly.

No. Because the ARP Reply is a unicast frame destined to A, and A and B are in the same LAN (Subnet 1), the frame is normally switched within Subnet 1 and does not traverse the link toward S1.

P18

At $t = 0$, A starts transmitting. The one-way propagation delay between A and B is 325 bit-times, and in the worst scenario, A sends a minimum-sized Ethernet frame lasting $512 + 64 = 576$ bit-times, so A finishes at $t = 576$. To maximize the detection delay at A, let B begin transmitting just before A's first bit reaches B, e.g., at $t = 324$. Then B's first bit reaches A at $t = 324 + 325 = 649$ bit-times. Since $649 > 576$, A completes transmission before receiving B's signal and thus can fail to detect the collision in time, incorrectly assuming the frame was transmitted successfully.

P31

The host first runs DHCP to obtain its network configuration. It sends a DHCP Discover message inside an IP datagram addressed to 255.255.255.255, encapsulates that datagram in an Ethernet frame, and broadcasts it on the LAN. After the DHCP exchange completes, the host is assigned an IP address with a lease time, and it also learns the subnet mask, the default gateway (one or more first-hop router IP addresses), and the IP addresses of DNS servers. Because the ARP cache is initially empty, the host uses ARP to resolve the MAC address of the next hop on the local link (typically the default gateway; if the DNS server is on the same LAN, it may ARP for the DNS server's MAC as well). The browser then issues a DNS query to the configured DNS server to obtain the Web server's IP address; if the DNS server does not already have the mapping cached, it performs DNS resolution and returns the result. With the destination IP known, the host establishes a TCP connection to the server and sends an HTTP request over that connection. The HTTP message is carried in TCP segments, encapsulated in IP packets, and then placed into Ethernet frames addressed to the default gateway's MAC (unless the server is on the local subnet). The gateway router removes the Ethernet header, forwards the IP packets according to its routing table, and the packets traverse the Internet hop by hop to the Web server. The server returns the content in HTTP response messages over the same TCP connection, and the response packets are routed back to the first-hop router, which encapsulates them in Ethernet frames and delivers them to the host.